

2024 - OHIKOA

A3

Izan bedi $f(x) = \frac{x}{x^2 - 2x + 1}$. $f(x) = \frac{x}{(x-1)^2}$

a) ASINTOTAK

Dom $f = \mathbb{R} - \{1\}$

ASINT. BERTIKALA

Fu. arazional batean izendatzen berdin zero daukagu.

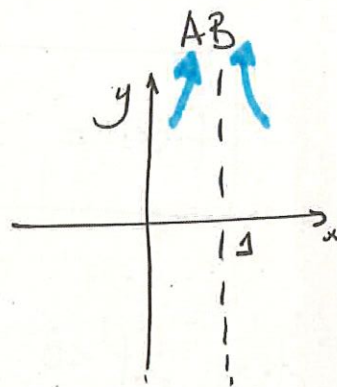
$\lim_{x \rightarrow a} f(x) = \pm \infty$

A.B $\boxed{x=1}$ $\lim_{x \rightarrow 1} f(x) = \left(\frac{1}{0}\right)$

$\frac{\rightarrow}{0.99} \quad 1 \quad \frac{\leftarrow}{1.01}$

$\lim_{x \rightarrow 1^-} \frac{x}{(x-1)^2} = \frac{1}{0^+} = +\infty$

$\lim_{x \rightarrow 1^+} \frac{x}{(x-1)^2} = \frac{1}{0^+} = +\infty$



A. HORIZONTALA

$\lim_{x \rightarrow \infty} f(x)$, fu arazional honetan zerbaitzaileroren uala, zerbaitzaileroren bikoizteagatik daukagu $\lim_{x \rightarrow \infty} f(x) = 0$, beraz $\boxed{y=0}$ da A.H.H.

Jokutako kurba asintototetikiko uala kokatzen dugu:

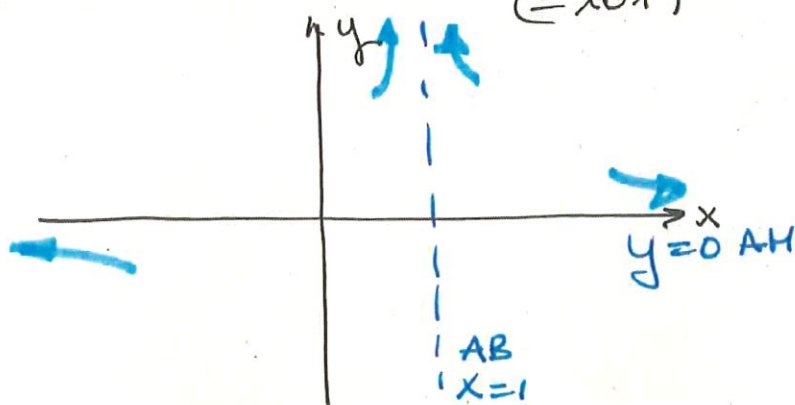
$x \rightarrow +\infty$ $f(100) = \frac{100}{99^2} = 0.01 > 0$

KURBA SAINETIK

$x \rightarrow -\infty$

$f(-100) = \frac{-100}{(-101)^2} = -0.01 < 0$

KURBA AZPIÑIK



b) GORAKORT. eto BETERAK. TARTEAK.

Dom $f = \mathbb{R} - \{1\}$

$$f(x) = \frac{x}{(x-1)^2} \quad f(x) = \frac{x}{x^2 - 2x + 1}$$

$$f'(x) = \frac{1 \cdot (x^2 - 2x + 1) - x(2x - 2)}{(x-1)^4} = \frac{x^2 - 2x + 1 - 2x^2 + 2x}{(x-1)^4}$$

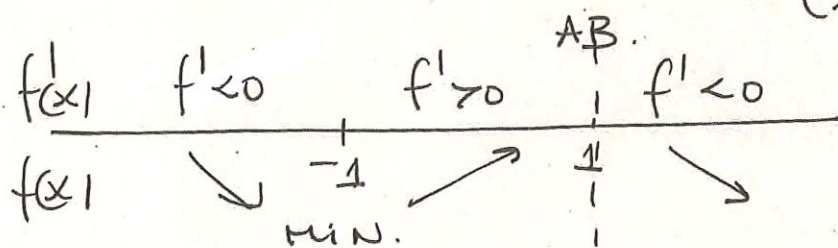
$$f'(x) = \frac{-x^2 + 1}{(x-1)^4} = \frac{(1+x)(1-x)}{(x-1)^4} = \frac{-(1+x)(x-1)}{(x-1)^4} =$$

$$f'(x) = \frac{-(1+x)}{(x-1)^3}$$

$$f'(x) = 0 \Rightarrow \frac{-(1+x)}{(x-1)^3} = 0$$

$$-(1+x) = 0 \rightarrow \boxed{x = -1}$$

ptu singular.



$$f(-1) = \frac{-1}{(-1-1)^2} = -1/4$$

$$f'(-2) = \frac{1}{-8} \ominus$$

$$f'(0) = \frac{-1}{-1} = \oplus$$

$$f'(2) = \frac{-3}{1} = \ominus$$

$$\begin{aligned} &G.T. (-1, 1) \\ &B.T. (-\infty, -1) \cup (1, +\infty) \\ &min. EXLAMBDA (-1, -1/4) \end{aligned}$$

c) UKITAILUA $x=0$ dauen.

Zuzen ukitzoileoren ekuazioa $y = y_0 + m(x - x_0)$ puntu bat eto molda behar da.

$$x=0 \rightarrow f(0) = \frac{0}{(0-1)^2} = 0 \rightarrow P(0, 0)$$

$$m = f'(x_0) \rightarrow m = f'(0) = \frac{-(1+0)}{(0-1)^3} = 1$$

$y = 0 + 1(x - 0)$

$$\boxed{y = x}$$

d) GRAFICA

$$f(x) = \frac{x}{(x-1)^2}$$

$$\text{Dom } f = \mathbb{R} - \{1\}$$

• Ebakitze puntuak

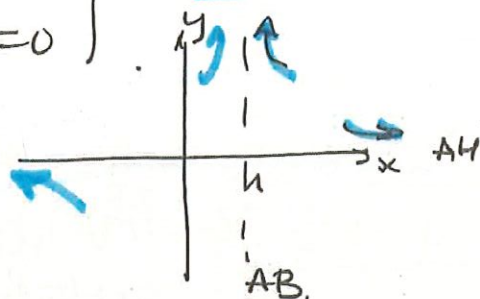
$$\left. \begin{array}{l} \text{Ox ardatz} \rightarrow y=0 \rightarrow x=0 \\ \text{Oy ardatz} \rightarrow x=0 \rightarrow y=0 \end{array} \right\}$$

P(0,0)

• Asintotok

AB $x=1$

AH $y=0$



• Monotonis

GT $(-1, 1)$

BT $(-\infty, -1) \cup (1, +\infty)$

Nin $(-1, -1/4)$

• Kurbadura

AHURNA $(-2, 1) \cup (1, +\infty)$

Gaurako $(-\infty, -2)$

Iuf. punt $(-2, -2/9)$

$$f(-2) = \frac{-2}{(-2-1)^2} = \frac{-2}{9}$$

KURBADURA.

$$f''(x) = 0 \text{ IP.}$$

$$f''(x) > 0 \cup$$

$$f''(x) < 0 \cap.$$

$$f''(x) = \frac{-(x-1)^3 - (-1-x)(x-1)^2 \cdot 3}{(x-1)^6} = \frac{(x-1)^2 (- (x-1) + 3(1+x))}{(x-1)^6}$$

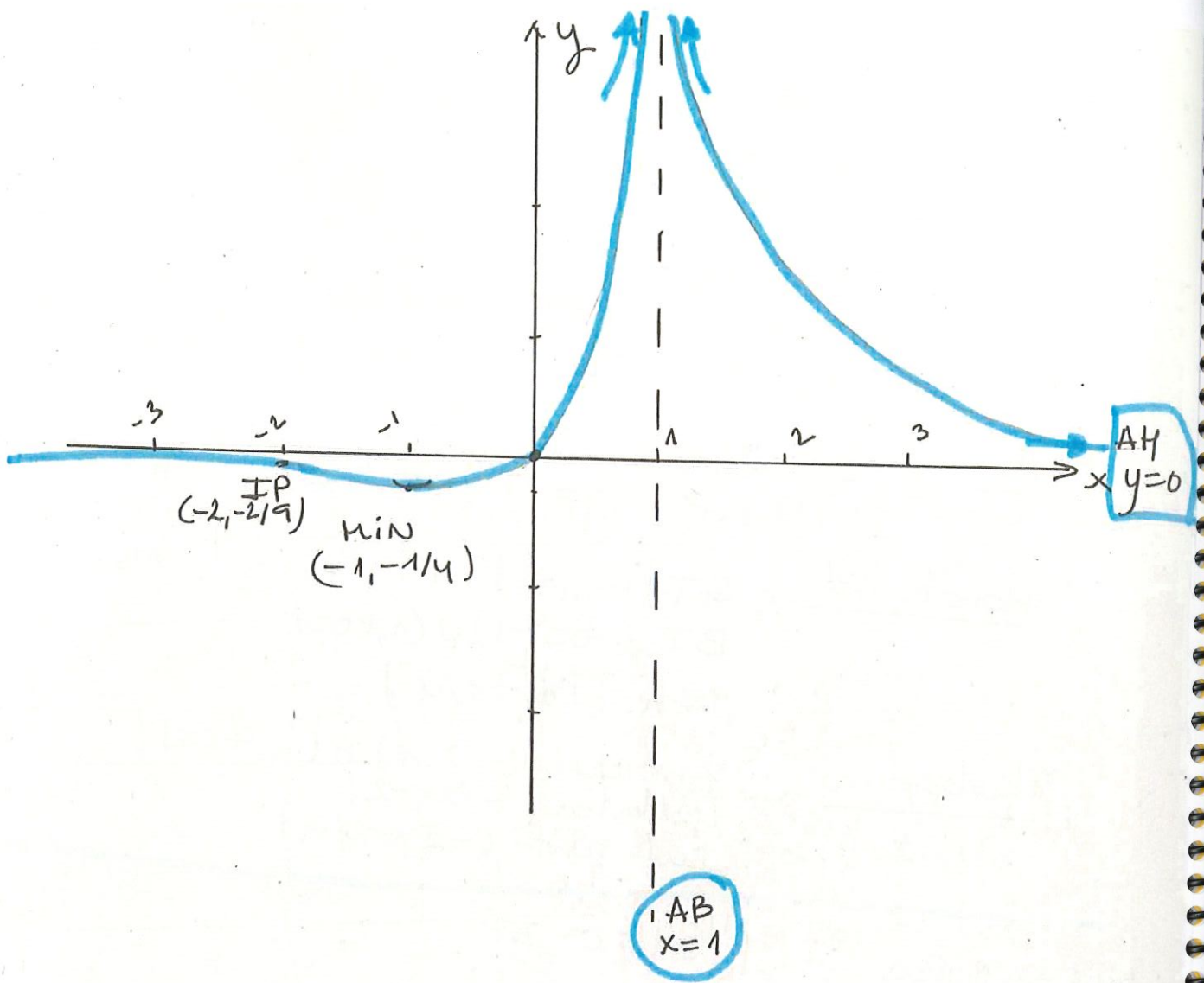
$$= \frac{-x+1+3+3x}{(x-1)^4} = \frac{2x+4}{(x-1)^4} = \frac{2(x+2)}{(x-1)^4}$$

$$f''(x) = 0 \rightarrow \frac{2(x+2)}{(x-1)^4} = 0 \rightarrow x = -2 \text{ Iuf. punt. baldin } f'''(x) \neq 0.$$

$$f''(x) \quad f'' < 0 \quad f'' > 0 \quad | \quad f'' > 0$$

$$f(x) \quad \cap \quad -2 \quad \cup \quad | \quad \cup$$

AB



B3 2024 OHIKOA

a) $f(x) = Ax^4 + Bx^2 + C$

• Mutur erlatiboa $x=1/2$

Mutur erlatiboetan $f'(x)=0$ beraz $f'(1/2)=0$

• Zuzen ukitzailearen ekuazioa $x=1$ denean $y=6x-2$

Zuzen ukitzailearen maldak $m=6$ da eta

bat dator $f'(1) \rightarrow m=f'(1) \rightarrow f'(1)=6$

Kurba eta ukitzailea bat dator

$x=1$ denean beraz:

$$x=1 \rightarrow y=6 \cdot 1 - 2 = 4 \rightarrow$$

$$f(1)=4$$

$$f(x) = Ax^4 + Bx^2 + C$$

$$f'(x) = 4Ax^3 + 2Bx$$

$$f(1)=4 \rightarrow A \cdot 1^4 + B \cdot 1^2 + C = 4 \rightarrow A + B + C = 4$$

$$f'(1/2)=0 \rightarrow 4A\left(\frac{1}{2}\right)^3 + 2B \cdot \frac{1}{2} = 0 \rightarrow \frac{A}{2} + B = 0$$

$$f'(1)=6 \rightarrow 4A \cdot 1^3 + 2B \cdot 1 = 6 \rightarrow 4A + 2B = 6$$

$$\begin{cases} A + 2B = 0 \end{cases}$$

$$\begin{cases} 4A + 2B = 6 \end{cases}$$

$$\begin{array}{r} -3A = -6 \rightarrow A = 2 \rightarrow B = -1 \rightarrow C = 3 \end{array}$$

Beraz:

$$f(x) = 2x^4 - x^2 + 3$$

b) Titik ekstrem

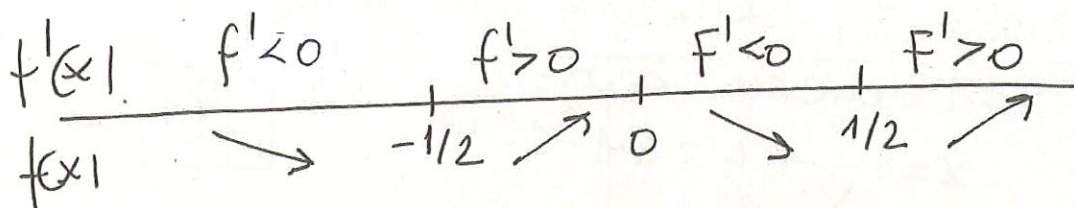
$$f(x) = 2x^4 - x^2 + 3$$

$$f'(x) = 8x^3 - 2x$$

Ptitik stingular $f'(x) = 0$.

$$0 = 8x^3 - 2x$$

$$2x(4x^2 - 1) = 0 \begin{cases} 2x = 0 \rightarrow x = 0 \\ 4x^2 - 1 = 0 \rightarrow x = \pm 1/2 \end{cases}$$



$$f(-1/2) = 2 \cdot (-0.5)^4 - (-0.5)^2 + 3 = 2.875$$

$$f(0) = 2 \cdot 0^4 - 0^2 + 3 = 3$$

$$f(1/2) = 2.875$$

max. ekstrem	$(-0.5; 2.875), (0.5; 2.875)$
min. ekstrem	$(0, 3)$

2024. ET OHIKOA

A3 $f(x) = x^3 + Ax^2 + Bx + C$

- zureu ukitzarleak $x = -1$ eta $x = 2$ -rako paraleloak

Paraleloak badira molda berdinak daukie, beraz $f'(x_0)$ bordinak daukie. $\rightarrow \boxed{f'(-1) = f'(2)}$

- $x = 1$ denean MUTUR ERLANBOA dauko.

Mutur erlatibetara $f'(x_0) = 0$ beraz:

$$\boxed{f'(1) = 0}$$

- $f(0) = \lim_{x \rightarrow 0} \frac{e^{2x} - 1}{x} = \left(\frac{0}{0}\right) \xrightarrow{\text{L'Hopital}} \lim_{x \rightarrow 0} \frac{2 \cdot e^{2x}}{1} = 2.$

Beraz $\boxed{f(0) = 2}$

Kalkulatuko da $f'(x)$ eta emandako baldintzak ordetketa:

$$f(x) = x^3 + Ax^2 + Bx + C$$

$$f'(x) = 3x^2 + 2Ax + B$$

$$f'(-1) = f'(2) \rightarrow 3 - 2A + B = 12 + 4A + B$$

$$-9 = 6A \rightarrow A = -\frac{9}{6} \rightarrow \boxed{A = -\frac{3}{2}}$$

$$f'(1) = 0 \rightarrow 3 + 2A + B = 0 \rightarrow 3 + 2 \cdot \left(-\frac{3}{2}\right) + B = 0$$

$$f(0) = 2 \rightarrow 0 + A \cdot 0 + B \cdot 0 + C = 2 \rightarrow \boxed{C = 2}$$

b) fren grafikoreu zureu ukitzarlearen ek.

$x = -1$ quen puntu etorko, $A = -3$, $B = 0$ eta

$C = 4$ -rako.

$f(x) = x^3 - 3x^2 + 4$. Izanda, zuren ukitzailearen
ekuazioa $y = y_0 + m(x - x_0)$ da.

$$\left. \begin{array}{l} y_0 = f(-1) \\ m = f'(-1) \end{array} \right\} \text{ lortuko da.}$$

$$y_0 = f(-1) = (-1)^3 - 3(-1)^2 + 4 = 0 \Rightarrow \boxed{y_0 = 0}$$

$$f'(x) = 3x^2 - 6x$$

$$f'(-1) = 3(-1)^2 - 6(-1) = 9 \longrightarrow \boxed{m = 9}$$

Beraz:

$$y = 0 + 9(x - (-1))$$

$$\boxed{y = 9x + 9}$$

B3 $f(x) = 2x e^{-2x^2}$ Domain = $\mathbb{R}$.

2024
E2 OHIKO

a) GORAKOR etc BEHERAKORTASUN TARTETAK.

$$f'(x) = 2 \cdot e^{-2x^2} + 2x \cdot (-4x) e^{-2x^2}$$

$$= e^{-2x^2} (2 - 8x^2)$$

$$f'(x) = 0 \rightarrow e^{-2x^2} (2 - 8x^2) = 0 \quad \left\{ \begin{array}{l} e^{-2x^2} \neq 0 \\ 2 - 8x^2 = 0 \end{array} \right.$$

$$\boxed{x = \pm \frac{1}{2}}$$

$f'(x)$	$f'(x) < 0$	$f'(x) > 0$	$f'(x) < 0$
$f(x)$	$\searrow$	$\nearrow$	$\searrow$
	$-\frac{1}{2}$	$\frac{1}{2}$	

G.T. $(-\frac{1}{2}, \frac{1}{2})$

B.T. $(-\infty, -\frac{1}{2}) \cup (\frac{1}{2}, +\infty)$

b) KUTUR ERLATIBOAK

Maximuetan $f''(x_0) < 0$

Minimuetan $f''(x_0) > 0$.

$$f''(x) = -4x e^{-2x^2} (2 - 8x^2) + e^{-2x^2} (-16x)$$

$$f''(x) = e^{-2x^2} (-8x + 32x^3 - 16x)$$

$$\boxed{f''(x) = e^{-2x^2} (32x^3 - 24x)}$$

$$\left. \begin{array}{l} f''(\frac{1}{2}) = e^{-2 \cdot \frac{1}{4}} (32(\frac{1}{2})^3 - 24 \cdot \frac{1}{2}) = \frac{1}{\sqrt{e}} (4 - 12) = -\frac{8}{\sqrt{e}} \text{ Max.} \\ f(\frac{1}{2}) = 2 \cdot \frac{1}{2} e^{-2(\frac{1}{2})^2} = e^{-1/2} \end{array} \right\}$$

$$\left. \begin{array}{l} f''(-\frac{1}{2}) = e^{-2 \cdot \frac{1}{4}} (-32(\frac{1}{2})^3 + \frac{24}{2}) = \frac{1}{\sqrt{e}} (-4 + 12) = \frac{8}{\sqrt{e}} \text{ Min.} \\ f(-\frac{1}{2}) = e^{-2 \cdot \frac{1}{4}} \cdot 2(-\frac{1}{2}) = -\frac{1}{\sqrt{e}} \end{array} \right\}$$

Maximua. $(\frac{1}{2}, e^{-1/2})$ puntuan $f'(\frac{1}{2}) < 0$ dabo

Minimua $(-\frac{1}{2}, -\frac{1}{\sqrt{e}})$ puntuan $f'(-\frac{1}{2}) > 0$ dabo

c) aurkitu f -re asintotak

$\text{Dom } f = \mathbb{R}$ beroz ez dauko asintotz bethelik.

$\lim_{x \rightarrow \infty} f(x)$ kalkulatu

$$\Rightarrow \lim_{x \rightarrow +\infty} 2x \cdot e^{-2x} = \lim_{x \rightarrow +\infty} \frac{2x}{e^{2x}} = \frac{+\infty}{+\infty} = 0$$

Infinitok konparatu $y = e^{2x}$ reu infinitoko
malo gertoko infinitoko da $y = 2x$ reu
baino.

Beraz $x \rightarrow +\infty$ denean ASINTOTA HORIZONTALA

$$\Rightarrow \lim_{x \rightarrow -\infty} 2x \cdot e^{-2x} = \lim_{x \rightarrow +\infty} 2(-x) e^{-2(-x)^2} =$$

$$= \lim_{x \rightarrow +\infty} \frac{-2x}{e^{2x}} = \frac{-\infty}{+\infty} = 0 \rightarrow \text{ASINT. HORIZ. } y=0.$$

Ondorioz: Ez dauko asint. bethelik
baino bai asint. horizontale $y=0$.